More precisely this relation should be interpreted in the variational sense, so that we have, for any suitable test function ϕ, e.g. $\phi \in C^\infty$,

$$\int_{\mathcal{D}} \phi \rho(\boldsymbol{q},\boldsymbol{p},t)\mathrm{d}\omega = \int_{\mathcal{D}} \phi e^{t\mathcal{L}_{\mathrm{LD}}^\dagger}\rho_0(\boldsymbol{q},\boldsymbol{p})\mathrm{d}\omega.$$

This is of course an inner product, and due to the properties of the inner product, we may interpret the right hand side as

$$\int_{\mathcal{D}} \left[e^{t\mathcal{L}_{\mathrm{LD}}}\phi(\boldsymbol{q},\boldsymbol{p})\right]\rho_0(\boldsymbol{q},\boldsymbol{p})\mathrm{d}\omega,$$

which says that the average of a function ϕ with respect to the distribution at time t is equivalent to the average of the forward propagation of ϕ with respect to the initial distribution.

How rapidly an average converges will depend on the spectral properties of the operator as well as the choice of the initial distribution. For example, if we assume that $\lambda_1, \rho^{(1)}$ is some eigenvalue, eigenfunction pair with $\mathrm{Re}(\lambda_1) < 0$ and take the initial distribution to be

$$\rho_0 = \rho_{\mathrm{eq}} + \alpha\rho^{(1)},$$

for $\alpha \in \mathbb{R}$, then we have

$$\begin{aligned}\rho(\boldsymbol{q},\boldsymbol{p},t) = e^{t\mathcal{L}_{\mathrm{LD}}^\dagger}\rho_0 &= e^{t\mathcal{L}_{\mathrm{LD}}^\dagger}\rho_{\mathrm{eq}} + \alpha e^{t\mathcal{L}_{\mathrm{LD}}^\dagger}\rho^{(1)} \\ &= \rho_{\mathrm{eq}} + \alpha e^{t\lambda_1}\rho^{(1)}.\end{aligned}$$

Since all the eigenvalues lie to the left of a vertical line in the left half plane, for any ϕ in C^∞ we have

$$\int_{\mathcal{D}} \phi\rho(\boldsymbol{q},\boldsymbol{p},t)\mathrm{d}\omega = \mathrm{Av}_\beta(\phi) + \alpha e^{t\lambda_1}\int_{\mathcal{D}} \phi\rho^{(1)}\mathrm{d}\omega.$$

The latter integral is bounded since both ϕ and $\rho^{(1)}$ are in L^2, and since $\mathrm{Re}(\lambda_1) < 0$, we see that this converges exponentially to the desired average.